

6.a) A group of six months chickens reared on a high protein diet weight 1.2, 1.5, 1.1, 1.6, 1.4^(1.2) and 1.6 kg. A second group of five chickens, similarly treated except that they receive a low protein weight: 0.1, 1.0, 1.4, 1.0 and 1.3 kg. Test whether there is significant evidence that high protein diet has increased the weight of the chickens.

solⁿ Here,

for first group

Let X_1 = weight of chickens.

X_1	X_1^2
1.2	1.44
1.5	2.25
1.1	1.21
1.6	2.56
1.4	1.96
1.4	1.96
1.6	2.56
9.8	13.94

Sample size, $n_1 = 7$

Sample mean, $\bar{X}_1 = \frac{9.8}{7} = 1.4$

$$\text{Sample S.D, } s_1 = \sqrt{\frac{\sum X_1^2}{n} - \left(\frac{\sum X_1}{n}\right)^2}$$
$$= \sqrt{\frac{13.94}{7} - \left(\frac{9.8}{7}\right)^2} = 0.177$$

for second group

Let X_2 = weight of chickens

X_2	X_2^2
0.1	0.01
1.0	1.00
1.4	1.96
1.0	1.00
1.3	1.69
4.8	5.66

Sample size, $n_2 = 5$

Sample mean, $\bar{X}_2 = \frac{4.8}{5} = 0.96$

$$\text{Sample S.D, } s_2 = \sqrt{\frac{\sum X_2^2}{n} - \left(\frac{\sum X_2}{n}\right)^2}$$
$$= \sqrt{\frac{5.66}{5} - \left(\frac{4.8}{5}\right)^2} = 0.458$$

Now,

$$S_b = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{7 \times (0.177)^2 + 5 \times (0.458)^2}{7 + 5 - 2}}$$

$$= 0.356$$

$$\therefore S.E.(\bar{X}) = S_b \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \left\{ 0.356 \right\} \sqrt{\frac{1}{7} + \frac{1}{5}}$$

$$= 0.208$$

* Testing Procedures:

i> Null Hypothesis, $H_0: \mu_1 = \mu_2$

i.e. there is no significance difference between the groups of chickens in giving diet.

Alternative Hypothesis, $H_1: \mu_1 > \mu_2$ (right tailed test)
i.e. there is significance difference between the groups of chickens in giving diet.

ii> Level of significance, $\alpha = 5\% = 0.05$ (commonly)

iii> Test statistic: Under H_0 ;

$$t_{cal} = \frac{\bar{X}_1 - \bar{X}_2}{S.E.(\bar{X})} = \frac{1.4 - 0.96}{0.208} = 2.115$$

iv) Tabulated value :

$$t_{tab} = t(\alpha, n_1 + n_2 - 2) = t(0.05, 10) = 1.812$$

v) Decision :

Here, $|t_{cal}| = 2.115$; $|t_{tab}| = 1.812$

$\therefore |t_{cal}| > |t_{tab}|$. So H_0 is rejected i.e.
 H_1 is accepted.

vi) Conclusion :

We concluded that there is significance between the groups of chickens in giving diet.

2022-Fall (Pre-Board, NAST)

5.a) In a sample of 600 high school students from a city, 400 are found to use dot pens. In one of 900 from a neighbouring city, 450 are found to use dot pens. Do the data indicate that the cities are significantly different with respect to the habit of using dot pens around the students? ($\alpha = 5\%$).

Sol Here

City	Neighboring City
$n_1 = 600$	$n_2 = 900$
$x_1 = 400$	$x_2 = 450$
$P_1 = \frac{x_1}{n_1} = \frac{400}{600}$	$P_2 = \frac{x_2}{n_2} = \frac{450}{900}$
$= 2/3$	$= 1/2$

We have,

$$\hat{p} = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} = \frac{600 \times \frac{2}{3} + 900 \times \frac{1}{2}}{600 + 900}$$
$$= 0.567$$

$$\therefore \hat{Q} = 1 - \hat{p} = 1 - 0.567 = 0.433$$

Now,

$$S.E.(\bar{X}) = \sqrt{\hat{p} \hat{Q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{0.567 \times 0.433 \left(\frac{1}{600} + \frac{1}{900} \right)}$$
$$= 0.0261.$$

* Testing Procedure:

i) Null Hypothesis, $H_0: P_1 = P_2$

i.e. there is no significance between two cities of using dot pens.

Alternative Hypothesis, $H_1: P_1 \neq P_2$ (two tailed test)
i.e. there is significance between the proportion of cities of using dot pens.

ii) Level of Significance: $\alpha = 5\% = 0.05$

iii) Test statistic: Under H_0 ,

$$Z_{cal} = \frac{P_1 - P_2}{S.E.(\bar{X})} = \frac{2/3 - 1/2}{0.0261} = 6.386$$

iv) Tabulated value:

$$Z_{tab} = Z_{\frac{0.05}{2}} = 1.96$$

$$\begin{aligned} & \frac{0.5 - 0.05}{2} \\ &= 0.475 \\ & (1.96) \end{aligned}$$

v) Decision:

$|Z_{cal}| > |Z_{tab}|$ i.e. H_0 is rejected.
 H_1 is accepted.

vi) Conclusion:

We concluded that there is significance between the proportions of two cities of using dot pens.

5.67 Twelve college students were given a test in statistics. They are given a month's tuition and a second test was held at the end of it. Do the marks give evidence that the students have benefited by the extra coaching? Use $\alpha = 0.05$.

Students	1	2	3	4	5	6	7	8	9	10	11	12
Marks in 1st test	23	20	19	21	18	20	18	17	23	16	19	24
Marks in 2nd test	24	19	22	18	20	22	20	20	23	20	18	22

Here,

Let X and Y be the marks in 1st test and marks in 2nd test respectively. Then,

X	Y	$d = X - Y$	d^2
23	24	-1	1
20	19	1	1
19	22	-3	9
21	18	3	9
18	20	-2	4
20	22	-2	4
18	20	-2	4
17	20	-3	9
23	23	0	0
16	20	-4	16
19	18	1	1
24	22	2	4
		-10	62

Now,

$$\bar{d} = \frac{\sum d}{n} = \frac{-10}{12} = -0.833$$

$$s_d = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} = \sqrt{\frac{62}{12} - \left(\frac{-10}{12}\right)^2}$$
$$= 2.115$$

Now,

$$S.E.(\bar{x}) = \frac{s_d}{\sqrt{n-1}} = \frac{2.115}{\sqrt{12-1}} = 0.6377$$

* Testing Procedures:

i) Null Hypothesis, $H_0: \mu_x = \mu_y$
i.e. there is no significance between the marks of students in 1st and 2nd test.

Alternative Hypothesis, $H_1: \mu_x < \mu_y$ (~~one~~ left tailed test)
i.e. there is significance between the marks of students in 1st and 2nd coaching test

ii) Level of significance $\alpha = 0.05$

iii) Test statistic: Under H_0 ,

$$t_{cal} = \frac{\bar{d}}{S.E.(\bar{x})} = \frac{-0.833}{0.6377} = -1.306$$

$$\therefore |t_{cal}| = 1.306$$

iv) Tabulated Value:

$$t_{tab} = t_{0.05, n-1} = t_{0.05, 11} = ~~1.645~~ 1.798$$

$$\text{i.e. } |t_{cal}| = ~~1.645~~ 1.798$$

v) Decision:

$$|t_{cal}| < |t_{tab}| \text{ i.e. } H_0 \text{ is accepted.}$$

vi) Conclusion:

We concluded that there is no significance between the ~~students~~ marks of students in 1st and 2nd tests.

2023-Spring

5. b) In a random sample of 400 students of the university teaching department, it was found that 300 students failed in the examination. In another random sample of 500 students of the affiliated colleges, the number of failures in the same examination was found to be 300. Find out whether the proportion of failures in the university teaching department is significantly greater than that of affiliated colleges. Use $\alpha = 0.05$.

solⁿ Here,

University teaching department	Affiliated colleges
$n_1 = 400$	$n_2 = 500$
$x_1 = 300$	$x_2 = 300$
$p_1 = 3/4$	$p_2 = 3/5$

we know,

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{400 \times 3/4 + 500 \times 3/5}{400 + 500}$$
$$= \frac{600}{900} = \frac{2}{3}$$

$$\hat{q} = 1 - \hat{p} = 1 - \frac{2}{3} = \frac{1}{3}$$

Now,

$$S.E(\bar{x}) = \sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{\frac{2}{3} \times \frac{1}{3} \left(\frac{1}{400} + \frac{1}{500}\right)}$$
$$= 0.0316$$

* Testing Procedure:

i> A Null Hypothesis, $H_0: P_1 = P_2$

i.e. there is no significance difference between two proportions of failures.

Alternative Hypothesis, $H_1: P_1 > P_2$ (right tailed test)

i.e. there is significance difference between two proportions of failures.

ii> Level of significance: $\alpha = 0.05$

iii> Test statistic; Under H_0 ;

$$Z_{cal} = \frac{P_1 - P_2}{S.E(\bar{x})} = \frac{3/4 - 3/5}{0.0316} = 4.747$$

iv> Tabulated value:

$$Z_{tab} = Z_{0.05} = 1.645$$

v> Decision:

$$|Z_{cal}| > |Z_{tab}|$$

so H_0 is rejected and H_1 is accepted.

vi> Conclusion:

We concluded that there is significance difference between two proportions of failures. This means the proportions of failures in the university teaching department is significantly greater than that of affiliated colleges.

6.a) An I.Q. test was administered to 5 persons before and after they were trained. The results given below:

Candidates	1	2	3	4	5
I.Q. before training	110	120	123	132	125
I.Q. after training	120	118	125	136	121

Is the training effective at 5% level of significance?

~~80~~

Let X_1 = I.Q. before training

X_2 = I.Q. after training

X	Y	$d = X - Y$	d^2
110	120	-10	100
120	118	2	4
123	125	-2	4
132	136	-4	16
125	121	4	16
		$\Sigma d = -10$	$\Sigma d^2 = 140$

We know,

$$\bar{d} = \frac{\Sigma d}{n} = \frac{-10}{5} = -2$$

$$S_d = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{140}{5} - \left(\frac{-10}{5}\right)^2} = 4.898$$

Now,

$$SE(\bar{X}) = \frac{S_d}{\sqrt{n-1}} = \frac{4.898}{\sqrt{5-1}} = 2.449$$

* Testing Procedure:

- i) Null Hypothesis, $H_0: \mu_x = \mu_y$
i.e. there is no significance between the I.Q. test of candidates before and after training.

Alternative Hypothesis, $H_1: \mu_x < \mu_y$ (left tailed test)
i.e. there is significance between the I.Q. test of candidates after before and after training.

ii) Level of significance: $\alpha = 5\% = 0.05$

iii) Test statistic: Under H_0 ,

$$t_{cal} = \frac{\bar{d}}{S.E(\bar{X})} = \frac{-2}{2.449} = -0.8167$$

iv) Tabulated value:

$$t_{tab} = t_{(0.05, 5-1)} = t_{(0.05, 4)} = 2.132$$

v) Decision:

$|t_{cal}| < |t_{tab}|$ i.e. H_0 is ~~rejection~~ accepted.

vi) conclusion:

We concluded that there is significance between the I.Q. test of candidates before and after training. This means the training ~~effective~~ is effective at 5% level of significance.

2022-Fall

5.6) In a factory 2% fans were found to be defectives in a lot of 3000 fans and in other factory 3% fans were found to be defective in a lot of 2500 fans. Do you find that the fans in secondary factory are significantly inferior compared to the fans in the first factory? (Use $\alpha = 5\%$).

solⁿ Here,

First factory	Second factory
$n_1 = 3000$	$n_2 = 2500$
$x_1 = 2\% \text{ of } 3000$ $= 60$	$x_2 = 3\% \text{ of } 2500$ $= 75$
$p_1 = \frac{60}{3000} = 0.02$	$p_2 = \frac{75}{2500} = 0.03$

We know,

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{3000 \times 0.02 + 2500 \times 0.03}{3000 + 2500}$$
$$= 0.02454$$

$$\therefore \hat{q} = 1 - \hat{p} = 1 - 0.02454 = 0.97546$$

Now,

$$S.E(\bar{x}) = \sqrt{\hat{p} \hat{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$
$$= \sqrt{0.02454 \times 0.97546 \left(\frac{1}{3000} + \frac{1}{2500} \right)}$$
$$= 0.004189$$

* Testing Procedures:

i) Null Hypothesis, $H_0: P_1 = P_2$

i.e. there is no significance difference between the proportions of two factories.

Alternative Hypothesis, $H_1: P_1 < P_2$ (left tailed test)
i.e. there is significance difference between the proportions of two factories.

ii) Level of significance: $\alpha = 5\% = 0.05$

iii) Test Statistic: Under H_0 ,

$$Z_{cal} = \frac{P_1 - P_2}{S.E(\bar{x})} = \frac{0.02 - 0.03}{0.004189} = -2.3872$$

$$\therefore |Z_{cal}| = 2.3872$$

iv) Tabulated Value:

$$Z_{tab} = Z_{0.05} = -1.645$$

$$\therefore |Z_{tab}| = 1.645$$

v) Decision:

$$|Z_{cal}| > |Z_{tab}| \text{ i.e. } H_0 \text{ is rejected.}$$

vi) Conclusion.

We concluded that there is significance between the proportions of two factories. This means the fans in 2nd factory are significantly inferior than 1st factory.

6. a) Resistors labeled as hundred ohm are purchased from two different vendors. The specification for this type of resistor is that its actual resistance be within 5% of its labeled resistance. In a sample of 180 resistors from vendor A, 150 of them met the specification. In a sample of 270 resistors purchased from vendor B, 233 of them met the specification. Vendor A is the current supplier, but if the data demonstrate convincingly that a greater proportion of the resistors from vendor B meet the specification, a change will be made.

- i) State null and alternate hypothesis.
- ii) Calculate test static and draw a conclusion that a change should be made? (use $\alpha = 5\%$).

80 Here,

For vendor A	For vendor B
$n_1 = 180$	$n_2 = 270$
$x_1 = 150$	$x_2 = 233$
$p_1 = \frac{x_1}{n_1} = \frac{150}{180} = \frac{5}{6}$	$p_2 = \frac{x_2}{n_2} = \frac{233}{270}$
$= 0.83$	$= 0.86$

We know that,

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{150 + 233}{180 + 270} = 0.851$$

$$\therefore \hat{Q} = 1 - \hat{p} = 1 - 0.851 = 0.149$$

Now,

$$S.E.(\bar{X}) = \sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{0.851 \times 0.149\left(\frac{1}{180} + \frac{1}{270}\right)}$$

$$\therefore S.E.(\bar{X}) = 0.0342$$

* Testing Procedure:

i) Null Hypothesis, $H_0: P_1 = P_2$

i.e. there is no diff significance between the two proportions of vendor A and vendor B.

Alternative Hypothesis, $H_1: P_1 < P_2$ (left tailed test)

i.e. there is significance between the two proportions of vendor A and vendor B.

ii) Level of significance: $\alpha = 5\% = 0.05$

iii) Test statistic: Under H_0 ;

$$Z_{cal} = \frac{P_1 - P_2}{S.E.(\bar{X})} = \frac{0.83 - 0.86}{0.0342} = -0.8772$$

$$\text{i.e. } |Z_{cal}| = 0.8772$$

iv) Tabulated value:

$$Z_{tab} = Z_{0.05} = -1.645$$

$$\therefore |Z_{tab}| = 1.645$$

v) Decision:

$$|Z_{\text{cal}}| < |Z_{\text{tab}}|$$

So, H_0 is accepted.

vi) Conclusion:

We concluded that that there is no significance between two proportions of vendor A and vendor B.

2021-Spring

5.6) A manufacture of gun powder has developed a new powder which is designed to produce a muzzle velocity equal to 3000 ft/sec. Seven shells are loaded with the charge and muzzle velocities measured. The resulting velocities are as follows: 3005, 2935, 2965, 2905, 2995 and 3905. Does these data present sufficient evidence to indicate that the average velocity differs from 3000 ft/sec?

Sol Here,

Let X = muzzle velocities:

X	X^2
3005	9030025
2935	8614225
2965	8791225
2905	8439025
2995	8970025
3905	15249025
18710	59093550

Population, mean, $\mu_0 = 3000$ ft/sec

Sample size, $n = 6$

sample mean, $\bar{X} = \frac{18710}{6} = 3118.33$ ft/sec

$$\text{Sample S.D., } S = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{59093550}{6} - \left(\frac{18710}{6}\right)^2} = 353.44 \text{ ft/sec}$$

Now,

$$S.E.(\bar{X}) = \frac{s}{\sqrt{n-1}} = \frac{353.44}{\sqrt{6-1}} = 158.06$$

* Testing procedures:

i) Null Hypothesis, $H_0: \mu = 3000$

i.e. there is no significance difference between the mean of muzzle velocities.

Alternative Hypothesis, $H_1: \mu \neq 3000$ (Two tailed test)

i.e. there is significance difference between the mean of muzzle velocities.

ii) Level of significance: $\alpha = 0.05$ (commonly)

iii) Test statistic: Under H_0 ,

$$t_{cal} = \frac{\bar{X} - \mu}{S.E.(\bar{X})} = \frac{3118.33 - 3000}{158.06} = 0.74863$$

iv) Tabulated Value:

$$t_{tab} = t_{(\frac{0.05}{2}, n-1)} = t_{(\frac{0.05}{2}, 5)} = 2.571$$

v) Decision:

$$|t_{cal}| = 0.74863 \quad ; \quad |t_{tab}| = 2.571$$

$\therefore |t_{cal}| < |t_{tab}|$. So, H_0 is accepted.

vi) Conclusion:

We concluded that there is no significance between the mean of muzzle velocities. This means there is no differs.